

ERMI Analytical Test Report

Client:

Marisa Flores

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Sampled By

Marisa Flores

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Project Name:

Marisa Flores

Sample Type:

Swiffer Kit

Status

Non Available

Site Address:

4 Ellsworth Ave , Apt. 32, Cambridge,
MA, 02139, US

Sample Location:

Cambridge , MA

Date of Sampling:

Dec 12, 2024

Date Samples Received:

Dec 23, 2024

Date of Report:

Jan 02, 2025

Reference No.

424367

P.O.

62683

EB Code**Class N°****Check N°**

The samples collected by the client were referred under the chain of custody to our laboratory for analysis and reporting. The samples received were labeled, and their condition on receipt was intact. Per our Terms & Conditions, this document and its contents are intended for the Addressee only and not for general publication without written consent.

Group 1: Water Damage Molds

Species	Q Level	SE/mg	
Aspergillus flavus/oryzae		23	*
Aspergillus fumigatus		43	*
Aspergillus niger		717	*
Aspergillus ochraceus		10	
Aspergillus penicillioides		1,731	*
Aspergillus restrictus		11	
Aspergillus sclerotiorum		6	
Aspergillus sydowii		436	*
Aspergillus unguis		167	**
Aspergillus versicolor		73	
Aureobasidium pullulans		6,421	*
Chaetomium globosum		3	
Cladosporium sphaerospermum		34	
Eurotium (Asp.) amstelodami		333	
Paecilomyces variotii		181	*
Penicillium brevicompactum		233	*
Penicillium corylophilum		77	*
Penicillium crustosum		78	*
Penicillium purpurogenum		3	
Penicillium Spinulosum		5	
Penicillium variable		28	
Scopulariopsis brevicaulis/fusca		4	
Scopulariopsis chartarum		7	
Stachybotrys chartarum		32	*
Trichoderma viride		62	*
Wallemia sebi		11	
Sum of logs G1		43.8	

Group 2: Common Indoor Molds		
Species	Q Level	SE/mg
Acremonium strictum		23
Alternaria alternata		30
Aspergillus ustus		36 *
Cladosporium cladosporioides1		3,211
Cladosporium cladosporioides2		51
Cladosporium herbarum		99
Epicoccum nigrum		914 *
Mucor amphibiorum		7
Penicillium chrysogenum		397 *
Rhizopus stolonifer		N.D
Sample Size	Sum of logs G2	18.0
5.0	Ermi Results = (G1-G2)	25.8

			Q Level
SE	= Spore Equivalents	()	Normal
SE/mg	= SE/miligrams of sample	(*)	10 fold higher than normal.
Logs	= Logarithms	(**)	100 fold higher than normal.
ND	= None Detected	(***)	1,000 fold higher than normal.
			 Q1
			 Q2
			 Q3
			 Q4

COMMENTS

The table shows the Spore Equivalent per milligram (SE/mg) detected for each of the 36 environmental molds analyzed.

The stars symbols on table above highlights the main molds (DNAs) detected in this report, which were selected based on their value being higher than ten fold (*), 100 fold (**) and 1,000 folds (***) of the geometric mean of the corresponding mold on the 2007 USA survey of molds.

ERMI score was developed by the US government for environmental mold safety (mold related asthma) and the score table is a general recommendation.

For patients with CIRS condition, in general, an ERMI score of 2 or less is considered safe. For more information please consult with your doctor and an Indoor environmental professional for the best advice on how to interpret the results.

The interpretation was made with reference to the following table:

Environmental Relative Moldiness Index (ERMI)	25.8
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Level	ERMI Values	Interpretation	Comment
Q 1	Less than - 4	Low Relative Moldiness Index	Further investigation is not needed to determine the sources of the mold.
Q 2	-4 to < 0	Low - Medium Relative	Further investigation may be needed to determine the sources of the mold if occupants have been reactive, sensitized, genetically predisposed or otherwise immuno-compromised.
Q 3	0 to < 5	Medium- High Relative	
Q 4	5 to < 20	High Relative Moldiness Index	Source and cause of mold should be determined and remediation undertaken, reducing the ERMI to levels below Q2.
	> 20	Very High Relative	

According to Vesper [9] ERMI Scores have a Standard Deviation (S.D.) of +/-3 and should be assessed with this in mind.

Further assessment was performed by calculating the HERTSMI-2 score from this data, which was found to be:

HERTSMI-2 Species		Spore E./mg	Weighting
Aspergillus penicillioides	1,731		10 *
Aspergillus versicolor	73		4
Chaetomium globosum	3		0
Stachybotrys chartarum	32		6 *
Wallemia sebi	11		0
Sample Size	5.0 mg	HERTSMI-2 Score =	20

HERTSMI-2 Color-coded interpretation

If 10 or below	In only 1.7% of cases, re-occupancy of building following mold remediation has led to relapse of CIRS-WDB symptoms
If between 11 to 15	Borderline. Further remediation and re-assessment is indicated
If greater than 15	Re-occupancy is ill-advised until further remediation and re-assessment are conclusive.

A spore equivalent may reflect the presence of any other fungal structures (i.e. mycelia) containing the same number of target genes as a spore.

Approved by



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Lab Manager
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Genetically close-related species may be detected in the indicator assay.

As reported	Includes
Eurotium (Asp.) amstelodami	E. chevalieri, E. herbariorum, E. rubrum and E. repens.
Penicillium spinulosum	P. glabrum, P. lividum, P. pupureus, and P. thomii.
Trichoderma viride	T. koningii and T. atroviride.
Aspergillus restrictus	A. caesillus and A. conicus.
Mucor amphibiorum	M. circinelloides, M. hiemalis, M. indicus, M.ucedo, M. racemosus, M. ramosissimus.
Rhizopus zygosporus	R. homothalicus, R. microsporus, R. oligosporus, R. oryzae.
Penicillium crustosum. (P. Group-2)	P. camembertii, P. commune, P. echinulatum, P. solitum.
Aspergillus niger	Known as Aspergillus brasiliensis
Scopulariopsis brevicaulis/fusca	Has been renamed as species of Microascus ¹⁰
Wallemia sebi	W. mellicola, W. canadensis ¹¹

REFERENCES

“WHO Guidelines for Indoor Air Quality – Dampness and Mould”, 2009 World Health Organization, Copenhagen, Denmark, ISBN 978 92 890 4168 3.

“Development of an Environmental Relative Mouldiness Index” Vesper S. et al, Occupational Env. Med. 2007,49:829-833.

Reference Geometric Mean data from J Occup Environ Hyg. 2021 January; 18(1): 35–41